Quadratics: Expanding & Factorising

Practice test 2 — answer sheet · 40 marks

Section A — Expanding double brackets

A1. $x^2 + 9x + 3x + 27$ (1) = **$x^2 + 12x + 27$** (1)

A2. $x^2 – 2x – 8x + 16$ (1) = **$x^2 – 10x + 16$** (1)

A3. $(2x + 5)(2x + 5) = 4x^2 + 10x + 10x + 25$ (1) = **$4x^2 + 20x + 25$** (1)

A4. They squared each term separately and missed the two middle terms: $(x + 4)^2$ means $(x + 4)(x + 4)$ (1) **$x^2 + 8x + 16$** (1)

Section B — Expanding three brackets

B1. $(x + 3)(x – 1) = x^2 + 2x – 3$ (1) $(x^2 + 2x – 3)(x – 4) = x^3 – 4x^2 + 2x^2 – 8x – 3x + 12$ (1) = **$x^3 – 2x^2 – 11x + 12$** (1)

B2. $(x + 2)^2 = x^2 + 4x + 4$ (1) $(x^2 + 4x + 4)(x + 2) = x^3 + 2x^2 + 4x^2 + 8x + 4x + 8$ (1) = **$x^3 + 6x^2 + 12x + 8$** (1)

Section C — Factorising $x^2 + bx + c$ and the difference of two squares

C1. Multiply to –24, add to 2: 6 and –4 (1) **$(x + 6)(x – 4)$** (1)

C2. Multiply to 36, add to –13: –4 and –9 (1) **$(x – 4)(x – 9)$** (1)

C3. $(3x)^2 – 4^2$ = **$(3x + 4)(3x – 4)$** (1)

C4. $3 \times a = 15$ so **$a = 5$** (1) $b = 3 + 5$, so **$b = 8$** (1)

Section D — Factorising $ax^2 + bx + c$

D1. $5 \times –6 = –30$; numbers: 15 and –2, so $5x^2 + 15x – 2x – 6 = 5x(x + 3) – 2(x + 3)$ (1) **$(5x – 2)(x + 3)$** (1)

D2. $6 \times 4 = 24$; numbers: –8 and –3, so $6x^2 – 8x – 3x + 4 = 2x(3x – 4) – 1(3x – 4)$ (1) **$(2x – 1)(3x – 4)$** (1)

D3. $3x(x^2 – 4)$ (1) = **$3x(x + 2)(x – 2)$** (1)

Section E — Using factorising

E1. $(2x – 5)(x + 3) = 0$ (1) $2x – 5 = 0$ or $x + 3 = 0$ (1) **$x = 2.5$ or $x = –3$** (1)

E2. Numerator $= (2x + 1)(x – 4)$ (1) denominator $= (x + 4)(x – 4)$ (1) cancel $(x – 4)$: **$(2x + 1) / (x + 4)$** (1)

Section F — Problem solving & proof

F1. (a) $(x + 4)^2 = x^2 + 8x + 16$ (1) so $x^2 + 8x + 5 =$ **$(x + 4)^2 – 11$** (1)
(b) **$(–4, –11)$** (1)

F2. Let the numbers be $2n + 1$ and $2n + 3$ (1) $(2n + 3)^2 – (2n + 1)^2 = (4n^2 + 12n + 9) – (4n^2 + 4n + 1) = 8n + 8$ (1) $= 8(n + 1)$, which is **a multiple of 8** (1)

F3. (1001 + 999)(1001 – 999) = 2000 × 2 = **4000** (1)

Marking note: the marks in brackets show where each mark is earned. Award method marks for a correct method even if there is a later slip. Accept answers in any equivalent form, e.g. terms in a different order, or brackets written the other way round: (x – 7)(x + 4) is the same as (x + 4)(x – 7). “Factorise fully” questions lose the last mark if a common factor is left inside a bracket.